

lar Integral. Envelope of the surfaces represented by
ed is known as Particular Integral. Envelope of the surfaces represented by
lar Integral. Envelope of the surfaces represented by
gral of the given partial differential equation is called its

2.9 m dm. 1 kg bul = 1.9 m
0.1 kg bul 1 kg bul 1 kg bul

✱ Equation of 1st Order & 1st Degree Date.....

- Exercise - 27 → Linear differential equation
- Exercise - 28 → Reducible to linear equation
- Exercise - 29 → Exact differential eqⁿ
- Exercise - 30 → Reducible to exact differential eqⁿ
- Exercise - 33 → C.F.
- Exercise - 34 → C.F + P.I.
- Exercise - 35 → G.F + P.I.

✱ Linear differential eqⁿ of 2nd Order

- Exercise - 36 → Homogeneous differential eqⁿ
- Exercise - 37 → Exact differential eqⁿ
- Exercise - 38 → Part of G.F known
- Exercise - 39 → Removal of first derivative or
change of dependent variable / Reduction
to normal form
- Exercise - 40 → change of the independent
variable

200%

for Uni. Exam
→

- Exercise - 41 → Method of variation of Parameters
- Ques → 10 No.

✱ Asymptotes

- Exercise - 1 → Asymptotes of the given eqⁿ = ?
- Exercise - 2 → Intersection of the curve and
its asymptotes
- Exercise - 4 → Radius of curvature
- Exercise - 5 → Radius of curvature at origin }
at a point / at any point

Date.....

Exercise ~ 6 $\rightarrow$ Point of Inflexion

☀ Partial differentiation ~~eq~~

Exercise ~ 7 $\rightarrow$ Euler's theorem

Exercise ~ 8 $\rightarrow$ Differentiate ~ ~ ~

Exercise ~ 10 $\rightarrow$ Maxima & Minima

Exercise ~ 11 $\rightarrow$ Trace of the curve

☀ Partial differential eqⁿ $\rightarrow$

Language method

Exercise ~ 16 $\rightarrow$ Partial differential eqⁿ by the elimination of arbitrary constant / arbitrary function

Exercise ~ 17 $\rightarrow$ complete Integral & single integral
 $\downarrow$
standard form $C.F. = ?$ $S.I. = ?$
 $\underbrace{\hspace{10em}}_1$
 $C.F., S.I. = ?$

100% Exercise ~ 18 $\rightarrow$ Charpit method

$\downarrow$
for uni. exam

quesⁿ $\rightarrow$ 10 No.

★ Linear diff. eq.ⁿ

$$\frac{dy}{dx} + Py = Q$$

$$I.f = e^{\int P dx}$$

$$\text{Solution: } y \times I.f = \int Q \times I.f dx + C$$

★ Reducible to Linear diff. eq.ⁿ

$$\frac{dy}{dx} + Py + y^n Q \rightarrow \text{Bernoulli's equation}$$

$$y^{-n} \frac{dy}{dx} + Py^{(-n+1)} = Q$$

$$y^{-n+1} = v \quad \text{d.w.r.t } x$$

put in (1)

$$\frac{dv}{dx} + (1-n)P.v = (1-n)Q$$

★ Exact diff. eq.ⁿ

$$Mdx + Ndy = 0$$

$$\left[\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x} \right] \rightarrow \text{Exact}$$

$$\text{Solution: } \int M dx + \int N dy = C$$

★ Reducible to Exact

$$Mdx + Ndy = 0 \quad \text{--- (1)}$$

$$\frac{\partial M}{\partial y} \neq \frac{\partial N}{\partial x} \rightarrow \text{Not Exact}$$

• Case-1: $I.f = \frac{1}{Mx + Ny}$ if $Mx + Ny \neq 0$

$$(1) \times I.f$$

$$M'dx + N'dy = 0, \quad \frac{\partial M'}{\partial y} = \frac{\partial N'}{\partial x}$$

Same

Exact

• Case-2: $M \rightarrow f(xy) \cdot y$

$$N \rightarrow f(xy) \cdot x$$

So, $I.f = \frac{1}{Mx - Ny}$ then (1) $\times I.f$

• Case-3: $\frac{1}{N} \left(\frac{\partial M}{\partial y} - \frac{\partial N}{\partial x} \right) = f(x)/\text{const.}$

So, $e^{\int f(x)/\text{const.} dx}$

• Case-4: $\frac{1}{M} \left(\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y} \right) = f(y)/\text{const.}$

So, $e^{\int f(y)/\text{const.} dy}$

• Case-5: General method

let $I.f = x^h y^k$

then, (1) $\times I.f$

On comparing coeff. of power of x and y we get two equations. Solve the two eq.ⁿ and find value of h and k .

★ Finding (C.F) complementary function

Case 1: All roots are diff.

$$C.F = C_1 e^{m_1 x} + C_2 e^{m_2 x} + \dots + C_n e^{m_n x}$$

Case 2: two roots are same

$$C.F = (C_1 + C_2 x) e^{m_1 x} + C_3 e^{m_3 x} + \dots$$

Case 3: Complex roots $m = \alpha \pm i\beta$

$$C.F = e^{\alpha x} (C_1 \cos \beta x + C_2 \sin \beta x)$$

Case 4: Roots are same $m = \alpha \pm \sqrt{\beta}$

$$C.F = e^{\alpha x} (C_1 \cosh \sqrt{\beta} x + C_2 \sinh \sqrt{\beta} x)$$

★ Finding (P.I) Particular Integral

Case 1: General case

$$P.I = \frac{1}{D-a} Q = e^{ax} \int e^{-ax} Q dx$$

$\rightarrow \sec ax, \tan ax, \operatorname{cosec} ax, \cot ax$

$$e^{iu} = \cos u + i \sin u, \quad e^{-iu} = \cos u - i \sin u$$

$$\cosh u = \frac{e^u + e^{-u}}{2}, \quad \sinh u = \frac{e^u - e^{-u}}{2}$$

Case-2: $Q = e^{ax}$

$$P.I = \frac{1}{f(D)} e^{ax} = \frac{1}{f(a)} e^{ax} \quad (\text{if } f(a) \neq 0)$$

if $f(a) = 0$ then $P.I = \frac{x^k}{k!} e^{ax}$

Case-3: $Q = \sin ax$ or $\cos ax$

$$P.I = \frac{1}{f(a^2)} \sin ax / \cos ax \quad (\text{if } f(-a^2) \neq 0)$$

if $f(-a^2) = 0$ $P.I = \frac{-x}{2a} \cos ax$ for $\sin ax$

$P.I = \frac{x}{2a} \sin ax$ for $\cos ax$

Case-4: $Q = x^m$

$$P.I = \frac{1}{f(D)} x^m = [f(D)]^{-1} x^m$$

Case-5: $e^{ax} \cdot v \quad v = f(x)$

$$P.I = \frac{1}{f(D)} e^{ax} \cdot v = e^{ax} \frac{1}{f(D+a)} v$$

Case-6: $Q = uv \quad v = f(x)$

$$P.I = \frac{1}{f(D)} u \cdot v$$

$$P.I = u \times \frac{1}{f(D)} v - \frac{f'(D)}{(f(D))^2} v$$

$$(1 \pm u)^{-1} = 1 \pm u + u^2 \pm u^3 \dots u^m$$

$$\frac{d^m u^m}{du^m} = m!$$

if $m = m$

$$\frac{d^m u^m}{du^m} = 0$$

if $m > m$

★ Homogeneous diff eqⁿ of 2nd order

$$a_0 u^2 \frac{d^2 y}{du^2} + a_1 u \frac{dy}{du} + a_2 y = 0$$

$$\text{let } u = e^z, \quad z = \log u$$

$$D = \frac{d}{dz} \quad u \frac{dy}{du} = D'y$$

$$D'(D'-1)y = 0$$

$$\rightarrow a_0(a+bu)^2 \frac{d^2 y}{du^2} + a_1(a+bu) \frac{dy}{du} + a_2 y = 0$$

$$\text{let } (a+bu) = e^z \quad z = \log(a+bu)$$

$$(a+bu) \frac{dy}{du} = b D'y$$

$$(a+bu)^2 \frac{d^2 y}{du^2} = b^2 D'(D'-1)y$$

★ Exact diff equation

$$P_0 \frac{dy}{dx} + P_1 \frac{dy}{dx} + P_2 y = Q \quad (1)$$

$$P_0 - P_1' + P_0'' = 0 \rightarrow \text{Exact}$$

$$\text{Solution: } P_0 \frac{dy}{dx} + (P_1 - P_0')y = \int Q dx + C$$

★ Reducible to Exact

$$P_0 - P_1' + P_0'' \neq 0$$

$$\text{I.f} = u^m \text{ then I.f} \times (1)$$

$$\text{Equation} \rightarrow \text{exact} \rightarrow m=?$$

★ Part of CF Known

$$\frac{d^2 y}{du^2} + P \frac{dy}{du} + Qy = R$$

$$y = U \times V$$

$$\frac{d^2 V}{du^2} + \left[\frac{2}{U} \frac{dU}{du} + P \right] \frac{dV}{du} = \frac{R}{U}$$

$$\text{let } \frac{dV}{du} = Z \quad \frac{d^2 V}{du^2} = \frac{dZ}{du}$$

$$\frac{dZ}{du} + \left[\frac{2}{U} \frac{dU}{du} + P \right] Z = \frac{R}{U}$$

Solve like linear diff. eqⁿ

$$1 + P + Q = 0 \quad y = e^u = u$$

$$1 - P + Q = 0 \quad e^{-u} = u$$

$$P + Qu = 0 \quad u = u$$

$$1 + \frac{P}{a} + \frac{Q}{a^2} = 0 \quad e^{au} = u$$

$$2 + 2Pu + Qu^2 = 0 \quad u^2 = u$$

$$\int e^{ax} \cos bx \, dx = \frac{e^{ax}}{a^2 + b^2} [a \cos bx + b \sin bx] + C$$

$$\int e^{-ax} \cos bx \, dx = \frac{e^{-ax}}{a^2 + b^2} [-a \cos bx + b \sin bx] + C$$

$$\int e^{ax} \sin bx \, dx = \frac{e^{ax}}{a^2 + b^2} [a \sin bx - b \cos bx] + C$$

$$\int e^{-ax} \sin bx \, dx = \frac{e^{-ax}}{a^2 + b^2} [-a \sin bx - b \cos bx] + C$$

★ Removal of 1st derivative or change of Reduction to Normal form

$$\frac{d^2 y}{du^2} + P \frac{dy}{du} + Qy = R$$

$$\text{we know: } \frac{d^2 v}{du^2} + Iv = \frac{R}{u}$$

$$I = Q - \frac{1}{2} \frac{dP}{du} - \frac{P^2}{4} \Rightarrow \text{const.} / \frac{\text{const.}}{u^2}$$

$$u = e^{-\frac{1}{2} \int P du}$$

★ Change of Independent variable

$$\frac{d^2 y}{du^2} + P \frac{dy}{du} + Qy = R$$

$$\frac{d^2 y}{dz^2} + P_1 \frac{dy}{dz} + Q_1 y = R_1$$

$$\text{where: } P_1 = \frac{d^2 z}{du^2} + P \frac{dz}{du} \Rightarrow \text{const.} / \frac{\text{const.}}{u^2}$$

$$Q_1 = \frac{Q}{\left(\frac{dz}{du}\right)^2} \quad R_1 = \frac{R}{\left(\frac{dz}{du}\right)^2} \rightarrow \text{value in the form of } z$$

★ Method of Variation of Parameters

$$\frac{d^2 y}{du^2} + P \frac{dy}{du} + Qy = R$$

$$\text{C.S: } y = Au + BV \quad (3) \quad u, v \in f(u)$$

$$\text{C.F} = C_1 f_1(u) + C_2 f_2(u)$$

$$y = Au + BV$$

$$u \frac{dA}{du} + v \frac{dB}{du} = 0 \quad (1)$$

$$\frac{du}{du} \times \frac{dA}{du} + \frac{dv}{du} \times \frac{dB}{du} = R \quad (2)$$

Solve eq (1) & (2), find value of A & B put in (3) we get C.S

★ Partial differentiation

→ Homogenous function

$$f(u, y) = u^n \phi\left(\frac{y}{u}\right)$$

$$f(y, u) = y^n \phi\left(\frac{u}{y}\right)$$

→ Euler's theorem

$$u \frac{\partial f}{\partial u} + y \frac{\partial f}{\partial y} = n f$$

→ Euler's theorem of 2nd order

$$u \frac{\partial^2 f}{\partial u^2} + y \frac{\partial^2 f}{\partial y^2} = m f \quad (1)$$

diff w.r.t u and y (P) the (2) & (3)

Mult. (2) x u + (3) x y, we get

$$u^2 \frac{\partial^2 f}{\partial u^2} + 2uy \frac{\partial^2 f}{\partial u \partial y} + y^2 \frac{\partial^2 f}{\partial y^2} = m(m-1)f$$

→ total diff. coeff.

$$u = f(u, y) \quad u = f_1(t) \quad y = f_2(t)$$

$$\frac{du}{dt} = \frac{\partial u}{\partial u} \cdot \frac{du}{dt} + \frac{\partial u}{\partial y} \cdot \frac{dy}{dt}$$

★ Maxima and Minima

$$y = f(x) \\ \frac{dy}{dx} = f'(x) \quad \frac{d^2y}{dx^2} = f''(x)$$

for max or min

$$\frac{dy}{dx} = 0$$

$$\frac{d^2y}{dx^2} = +ve \rightarrow \text{Minima}$$

$$\frac{d^2y}{dx^2} = -ve \rightarrow \text{Maxima}$$

★ If $f = f(x, y)$ at point (a, b)

$$\frac{\partial f}{\partial x} = 0 \quad \frac{\partial f}{\partial y} = 0 \quad \text{find max and min point}$$

$$A = \frac{\partial^2 f}{\partial x^2} \quad B = \frac{\partial^2 f}{\partial x \partial y} \quad C = \frac{\partial^2 f}{\partial y^2}$$

Case 1: $AC - B^2 > 0$

• max $\rightarrow A < 0$

• min $\rightarrow A > 0$

Case 2: $AC - B^2 < 0$

neither max nor min

Case 3: $AC - B^2 = 0$

further investigation is necessary in this case, the point is stationary

★ Lagrange's method of Undetermined Multipliers

$$u = f(x, y, z)$$

$$\phi(x, y, z) = 0$$

$$Z = f(x, y) \quad V = f + \lambda \phi$$

Multipliers

★ Asymptotes

$$\text{put } x=1 \quad y=m$$

$$C = - \frac{\phi_{m-1}(m)}{\phi'_m(m)}$$

$$C = \frac{0}{0} \quad C = \frac{1}{0} \quad C = \frac{0}{5} = 0$$

indeterminate infinity (fail) zero (fail) zero (pass)

$$\rightarrow y = mx + C$$

when it fails:

$$\bullet \text{ two value same: } m = m_1 = m_2 \\ C^2 \phi''_m(m) + \frac{C}{L} \phi'_{m-1}(m) + \phi'_{m-2}(m) = 0$$

$$\bullet \text{ 3 value same: } m = m_1 = m_2 = m_3 \\ \frac{C^3}{L^3} \phi'''_m(m) + \frac{C^2}{L^2} \phi''_{m-1}(m) + \frac{C}{L} \phi'_{m-2}(m) + \phi'_{m-3}(m) = 0$$

★ Curvature

→ Cartesian form: $y = f(x)$ relation b/w x and y

$$\text{Radius } f = \frac{[1 + (\frac{dy}{dx})^2]^{3/2}}{d^2y/dx^2}$$

→ Parametric form: $x = f(t)$ relⁿ b/w x & t
 $y = f(t)$ relⁿ b/w y & t

$$\text{radius } f = \frac{[\dot{x}^2 + \dot{y}^2]^{3/2}}{\dot{x}\ddot{y} - \dot{y}\ddot{x}} \quad \begin{aligned} \dot{x} &= \frac{dx}{dt} \\ \ddot{x} &= \frac{d^2x}{dt^2} \\ \dot{y} &= \frac{dy}{dt} \\ \ddot{y} &= \frac{d^2y}{dt^2} \end{aligned}$$

→ Intrinsic form:

$$f = \frac{ds}{d\psi} \quad \text{relⁿ b/w } s \text{ and } \psi$$

★ Curvature at the Origin

→ Maclaurin's theorem:

$$p = \frac{dy}{dx} \text{ at } x=0, y=0 \quad q = \frac{d^2y}{dx^2} \text{ at } x=0, y=0$$

$$f = \frac{(1 + p^2)^{3/2}}{q}$$

★ Partial differential equation

→ Lagrange's form: $Pp + Qq = R$

$$P, Q, R = f(x, y, z)$$

$$p = \frac{\partial z}{\partial x} \quad q = \frac{\partial z}{\partial y}$$

$$\frac{dx}{P} = \frac{dy}{Q} = \frac{dz}{R} \rightarrow \text{Lagrange's linear equation}$$

→ Standard form-1: $f(p, q) = 0$

$$C.S.: z = ax + by + c \\ p = \frac{\partial z}{\partial x} = a \quad q = \frac{\partial z}{\partial y} = b \quad [p=a], [q=b]$$

put in $f(a, b) = 0$

→ Standard form-2: $f(z, p, q) = 0$

$$\text{let } x = u + ay \quad [p = \frac{dz}{dx}] \quad [q = a \frac{dz}{dx}]$$

$$\text{put in: } f[z, \frac{dz}{dx}, a \frac{dz}{dx}] = 0$$

→ Standard form-3: $f_1(u, p) = f_2(y, q) = a$

$$p = \phi_1(u, a) \quad q = \phi_2(y, a)$$

$$C.S.: dz = p du + q dy$$

→ Standard form-4 (Clairaut's form)

$$z = pu + qy + f(p, q)$$

$$p = a \quad q = b$$

$$C.S.: z = ax + by + f(a, b)$$

as usual: S.I and Cx.I

★ CHARPIT'S Method

$$f(x, y, z, p, q) = 0 \quad \text{C.S.: } dz = p dx + q dy$$

$$\frac{dp}{\frac{\partial f}{\partial x} + p \frac{\partial f}{\partial z}} = \frac{dq}{\frac{\partial f}{\partial y} + q \frac{\partial f}{\partial z}} = \frac{dz}{-\frac{\partial f}{\partial p} - q \frac{\partial f}{\partial q}} = \frac{dx}{-\frac{\partial f}{\partial p}} = \frac{dy}{-\frac{\partial f}{\partial q}} = \frac{dF}{0}$$

★ Curve tracing

→ Point of Inflexion:

A point on a curve at which the curve changes from concavity to convexity or from convexity to concavity is called point of Inflexion for the curve

→ Test for Concavity and Convexity

$$y \frac{d^2 y}{dx^2} \rightarrow \begin{array}{l} +ve : \text{Convex} \\ -ve : \text{Concave} \end{array}$$

$$x \frac{d^2 x}{dy^2} \rightarrow \begin{array}{l} +ve : \text{Convex} \\ -ve : \text{Concave} \end{array}$$

→ Test for point of Inflexion

$$\frac{d^2 y}{dx^2} = 0 \quad \frac{d^3 y}{dx^3} \neq 0$$

→ Multiple point: A point on curve where more than 1 branches passes through

→ Double point: A point on curve through which 2 branches of a curve passes

- Node: if 2 branches through double point on a curve are real and tangent to them are not coincident

- Cusp: if 2 branches through a d.p on curve are real & tangent to them are coincident

- Conjugate: If there are no real points on curve in the neighbourhood of a point P on the curve

★ Trace the Curve

→ Symmetry:

- if $f(x, y) = f(x, -y)$ about x-axis
- $f(x, y) = f(-x, y)$ about y-axis
- $f(x, y) = f(-x, -y)$ about origin
- $f(x, y) = f(y, x)$ about line $y = x$
- $f(x, y) = f(-y, -x)$ about line $y = -x$

Polium $\rightarrow x^3 + y^3 = 3axy$

→ Origin: put $x=0, y=0$

if is satisfied then curve passes through origin

→ Tangent at Origin:

Equate the lowest degree term of eqⁿ to zero

→ Intersection at axes:

- at x-axis put $y=0$ get $x=a$
- at y-axis put $x=0$ get $y=a$

→ Asymptotes: as usually

- // to x-axis / // to y-axis and oblique asy. = x does not // to y
- if Asy. present then curve is open
- if Asy not present then curve is close

